

QGroundControl – SKYDROID C12 Integration Specification

Camera / Gimbal / Zoom / Tracking / Target GPS Estimation

Implementation baseline: supplied C12 TOP Protocol V1.1.6 + supplied UAV Camera-to-Target GPS Estimation report

Status: Proposed implementation specification for a custom QGroundControl camera/gimbal module.

1. Objective

Add a C12-specific camera and gimbal integration to QGroundControl that provides operator controls for visible/thermal video, digital zoom, gimbal yaw/pitch, center/home, absolute angles, tracking, C12 attitude feedback, and target GPS estimation. The implementation must keep the C12 TOP protocol and the separate C12 AI binary protocol as distinct interfaces.

2. Interface Boundaries

Interface	Transport	Purpose
C12 TOP control	UDP/5000	Zoom, PTZ, yaw/pitch speed, absolute gimbal angle, attitude, legacy tracking
C12 AI	UDP/1030	AI enable, tracking region, target result/ROI and AI state
C12 visible video	RTSP/554	Visible stream
C12 thermal video	RTSP/555	Thermal stream
QGC internal	C++ ↔ QML	Expose state, commands, video/overlay and target estimate

3. Operator Feature Matrix

Feature	C12 / AI interface	QGC behavior
Visible / Thermal	RTSP source selection; AI Video=0/1	Source selector; restart video; clear stale overlay
Zoom 1–4×	TOP DZM 01–04	Preset buttons
Zoom + / –	TOP DZM 0A / 0B	Increment/decrement buttons
Manual gimbal	TOP PTZ 00–05	Press/hold arrows; STOP on release
Yaw velocity	TOP GSY	Signed 0.5°/s units
Pitch velocity	TOP GSP	Signed 0.5°/s units
Yaw + pitch velocity	TOP GSM	Joystick/proportional control
Absolute yaw/pitch	TOP GAY/GAP/GAM	Go-to-angle control, ±90°
Attitude	TOP GAA/GAC	Display yaw/pitch; freshness
Legacy tracking	TOP SUM/GOT	Tracking state/point only where used
AI tracking region	AI CW1	Operator draws region; send 1280×720 coordinates
SDK target ROI	AI CW2 result	Display exact SDK X0/Y0/X1/Y1; do not calculate replacement ROI
Target GPS	Internal estimator	Estimate only when required inputs are fresh/valid

4. Target GPS Estimation Model

The supplied estimation report assumes the target is centered in the camera frame. The estimate uses UAV latitude/longitude/altitude, UAV heading, gimbal yaw/pitch, target elevation, and a 6-inch (0.1524 m) camera forward offset.

$$H = Alt_u - Alt_t$$
$$R = H / \tan(|P|)$$
$$B = (Hdg - Yaw + 360) \bmod 360$$
$$\Delta N = R \cos(B) + 0.1524 \cos(Hdg)$$
$$\Delta E = R \sin(B) + 0.1524 \sin(Hdg)$$
$$Lat_t = Lat_u + \Delta N / 111320$$
$$Lon_t = Lon_u + \Delta E / [111320 \cos(Lat_u)]$$

Conventions from the supplied report: UAV heading 0° North and increases clockwise; gimbal yaw 0° forward, positive counterclockwise/left, negative clockwise/right; gimbal pitch 0° horizontal, positive up, negative down. The report recommends radians for sin/cos/tan and notes that a full 3D attitude model is required when UAV roll/pitch is significant.

5. Estimator Preconditions and State

Input	Required behavior
UAV latitude/longitude	Fresh and valid
UAV altitude	Must have a declared reference; target altitude must be available or an explicit assumption used
UAV heading	Fresh and valid
Gimbal yaw/pitch	Fresh C12 GAC values
Target centered	Required by the supplied simplified model
Target source	Visible/thermal must match the AI result
Time alignment	Telemetry/video/AI skew must be within configured limit
Pitch singularity	Reject/invalid when P is near 0°
Terrain	Use target elevation when available; do not silently assume flat terrain for high-accuracy mode

6. Estimator Pseudocode

```
FUNCTION estimateTargetGPS(uav, gimbal, target):

    REQUIRE uav.latitude_valid
    REQUIRE uav.longitude_valid
    REQUIRE uav.altitude_valid
    REQUIRE uav.heading_valid
    REQUIRE gimbal.yaw_valid
    REQUIRE gimbal.pitch_valid
    REQUIRE target.centered
    REQUIRE target.source == activeCameraSource

    H = uav.altitude - target.groundAltitude

    IF ABS(gimbal.pitch) < MIN_PITCH_ANGLE:
    RETURN INVALID("Pitch too close to horizontal")

    R = H / tan(abs(gimbal.pitch))

    B = MOD(uav.heading - gimbal.yaw + 360.0, 360.0)

    dN = R * cos(B) + 0.1524 * cos(uav.heading)
    dE = R * sin(B) + 0.1524 * sin(uav.heading)

    lat = uav.latitude + dN / 111320.0
    lon = uav.longitude + dE / (111320.0 * cos(radians(uav.latitude)))

    RETURN {
    status: VALID,
    latitude: lat,
    longitude: lon,
    horizontalRange: R,
    bearing: B,
    dNorth: dN,
    dEast: dE
    }
```

7. QGC Architecture

```
QML Fly View / Camera UI
■
▼
C12CameraController
■■■ C12TopProtocolV116
■ ■■■ QUDpSocket : 5000
■■■ C12AIProtocolV120
■ ■■■ QUDpSocket : 1030
■■■ C12VideoManager
■ ■■■ RTSP visible : 554
■ ■■■ RTSP thermal : 555
■■■ TargetLocationEstimator
■■■ PX4 vehicle state
■■■ GAC gimbal state
■■■ AI SDK ROI/result
■■■ camera calibration / lever arm
```

8. Safety / Failure Handling

- Manual gimbal movement must transmit PTZ STOP when the operator releases the control or the module enters a safe-stop state.

- Loss of C12 control link stops new motion commands; attitude/target data become stale after their configured freshness timeout.
- Malformed C12 frames are rejected and logged.
- A stale AI ROI/result must not be used for target GPS estimation.
- The estimator must not output VALID when required altitude, heading, gimbal angle, or target-source inputs are invalid.
- Target GPS estimation is non-actuating in this feature; it reports a position and quality/status to QGC.
- Roll/pitch-rich 3D estimation should be a separate future enhancement because the supplied model is a simplified heading-only model.

9. Acceptance Criteria

1. QGC can select visible and thermal C12 streams.
2. Zoom actions generate the documented DZM values.
3. Manual yaw/pitch commands generate the documented GSY/GSP/GSM values and STOP is sent on release.
4. Absolute yaw/pitch commands enforce $\pm 90^\circ$ limits.
5. GAA/GAC attitude is displayed with documented 0.01° decoding.
6. AI tracking region is sent in C12 AI 1280×720 coordinates.
7. SDK target ROI is displayed exactly as reported by the SDK; no locally calculated replacement ROI is presented as SDK ROI.
8. Target GPS estimate matches the supplied mathematical model for the documented test cases.
9. Estimator returns INVALID for missing/stale inputs and near-horizontal pitch.
10. 6-inch / 0.1524 m lever arm is included in the north/east displacement.
11. All protocol operations have packet-level test evidence and UI acceptance tests.

10. Source Notes

The supplied ZIP contains the C12 interface contract/pseudocode and the UAV Camera-to-Target GPS Estimation report. The C12 protocol source defines UDP/5000, gimbal/zoom commands, attitude and legacy tracking. The estimation report defines the heading/yaw/pitch conventions, 6-inch lever arm and target GPS equations used above.